Sustainable Energy: Wind Generation Profiles and Solar Comparison

Joshan Rai
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1 Wind Generation Profile

1.1 Power Curve Approximation

To estimate the power output of the 10 MW SeaTitan wind turbine, we developed a piecewise mathematical function to approximate the manufacturer's power curve. The turbine operates across four distinct regimes based on hub-height wind speed (v): below cut-in, the ramp-up phase, the rated plateau, and above cut-out.

Based on the technical data, the parameters are:

- **Cut-in wind speed (v_{in}):** 4.0 m/s
- **Rated wind speed (v_{rated}):** 11.5 m/s
- **Cut-out wind speed (v_{out}):** 30.0 m/s
- **Rated Power (P_{rated}):** 10,000 kW

The power output function is defined as:

$$P(v) = \begin{cases} 0, & v < 4.0 \\ 10{,}000 \cdot \left(\dfrac{v - 4.0}{11.5 - 4.0}\right)^3, & 4.0 \leq v \leq 11.5 \\ 10{,}000, & 11.5 < v \leq 30.0 \\ 0, & v > 30.0 \end{cases} \tag{1}$$

1.2 Data Acquisition and Height Adjustment

Hourly wind speed data for the year 2020 at the Altamont Pass location was obtained using the National Laboratory of the Rockies (NLR, formerly NREL) WIND Toolkit Long-Term Ensemble Dataset (WTK-LED) API.

Because the WTK-LED dataset provides wind speeds at **100 meters**, while the SeaTitan has a hub height of **125 meters**, a height correction was applied. Using the power law profile, we adjusted the wind speeds using a shear exponent (α) of **0.10**, which is representative of the complex terrain in the Altamont Pass region.

$$v_{125} = v_{100} \cdot \left(\frac{125}{100}\right)^{0.10} \tag{2}$$

1.3 Hourly Power Conversion

Using the Python function defined in Section 1.1, we converted the 8,784 hourly wind speeds (adjusted to 125m for the leap year 2020) into hourly power output data. The resulting time series represents the theoretical generation profile of the turbine, accounting for zero-output periods when winds fall below 4.0 m/s or exceed 30.0 m/s.

1.4 Wind Generation Heat Map

The hourly power data was plotted on a heat map, with the hour of the day on the y-axis and the day of the year on the x-axis.

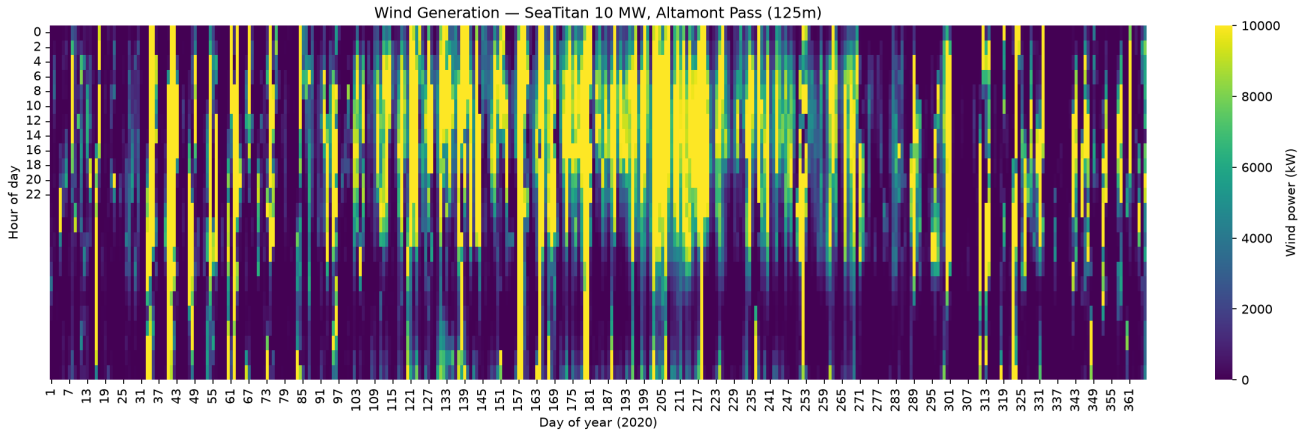


Figure 1: Wind Generation Heat Map for Altamont Pass (2020)

Observations: The heat map reveals significant hour-to-hour and day-to-day variability. However, there is a noticeable pattern of higher power generation occurring during the **night and early morning hours**, likely due to nocturnal boundary layer stabilization and the specific diurnal wind patterns of the Altamont Pass. Daytime generation is frequently lower or zero, especially during the summer months.

2 Solar Generation Profile

2.1 Data Acquisition via PVWatts

To contrast wind with solar, we calculated the hourly generation for a similar 10 MW solar array at the same location. We used the NLR PVWatts API (v8) with the following system parameters:

- **System Capacity:** 10,000 kW (10 MW DC)
- **Module Type:** Standard (0)
- **Array Type:** Fixed Open Rack (0)
- **Tilt:** 37° (optimal for latitude)
- **Azimuth:** 180° (Due South)
- **System Losses:** 14%

The API returned the hourly AC power output for a Typical Meteorological Year (TMY).

2.2 Solar Generation Heat Map

Similar to the wind data, the hourly solar power was plotted on a heat map to visualize the diurnal and seasonal generation patterns.

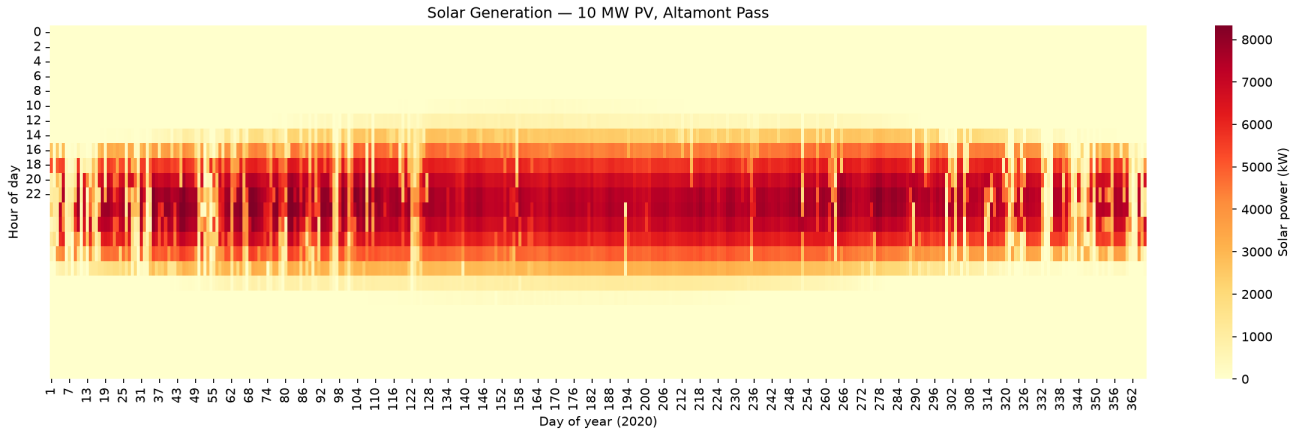


Figure 2: Solar Generation Heat Map for Altamont Pass

Observations: The solar heat map exhibits highly predictable, distinct behavior. Generation is strictly limited to **daylight hours**, peaking around solar noon. As expected, there is a massive seasonal variation, with summer days producing substantially more power over a longer duration than winter days. There is zero generation at night.

3 Comparative Analysis

3.1 Differences

- **Temporal Availability:** Solar generation is completely absent at night, whereas wind generation is often strongest at night or in the early morning.
- **Seasonality:** Solar output is highly deterministic and seasonal, driven by the earth's tilt and sun angle (peaking in summer). Wind output is stochastic and less predictable, driven by weather systems and local topography.
- **Variability:** The solar heat map shows a smooth, consistent "block" of generation centered around noon. The wind heat map is highly fragmented, showing day-to-day volatility driven by changing wind regimes.

3.2 Similarities and Complementarity

- **Rated Capacity Constraints:** Both systems are constrained by their 10 MW rated capacities. During peak solar noon in the summer, the PV array hits its maximum output, just as the wind turbine plateaus at 10 MW during high-wind events.
- **Variability:** Both are variable renewable energy (VRE) resources, meaning they cannot be dispatched on demand without energy storage or backup generation.
- **Resource Complementarity:** The most critical similarity is how the two resources complement each other. Solar peaks during the day, while wind frequently peaks at night. If combined with storage or transmission, a hybrid wind-solar system at this location could provide a much more stable and continuous power output than either resource alone.

References

1. Course Assignment Materials. (n.d.). *Sustainable Energy Wind technical assignment: 10 MW SeaTitan Power Curve*.

2. National Laboratory of the Rockies (NLR). (2024). *PVWatts API Version 8*. NLR Developer Network. `https://developer.nlr.gov/docs/solar/pvwatts/v8/`

3. National Laboratory of the Rockies (NLR). (2024). *WIND Toolkit Long-Term Ensemble Dataset (WTK-LED)*. NLR Developer Network. `https://developer.nlr.gov/docs/wind/wind-toolkit/`

4. Rai, J. (2026). *Wind and Solar Generation Profile Analysis Script*. `https://github.com/pradheon/wind-solar-analysis`